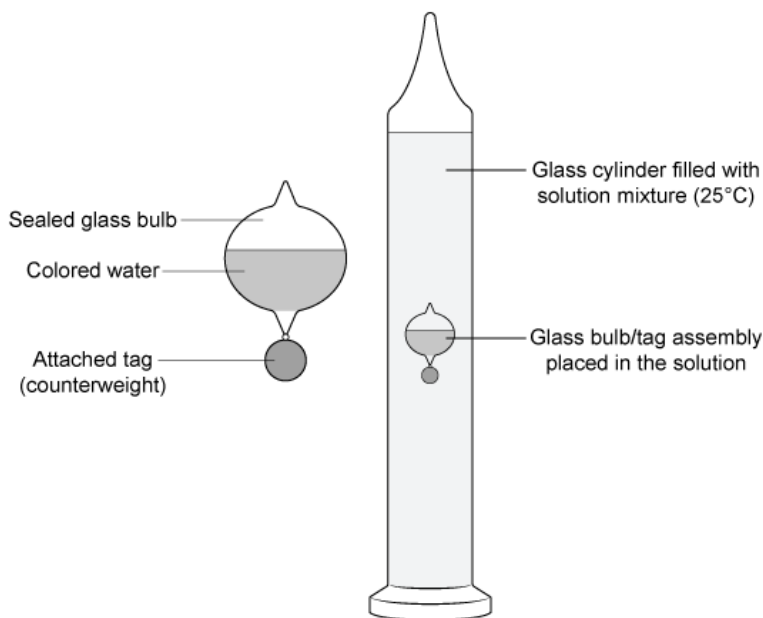


A Galileo thermometer consists of a cylinder filled with a variable solution mixture and several sealed glass bulbs partially filled with water. The density of the solution in the cylinder varies significantly with temperature, but the change in the density of the water inside each bulb is negligible. If one of the bulbs has the same density as the cylinder solution at 25 °C, how will the bulb's position and the density of the cylinder solution change as the temperature decreases from 25 °C to 20 °C?

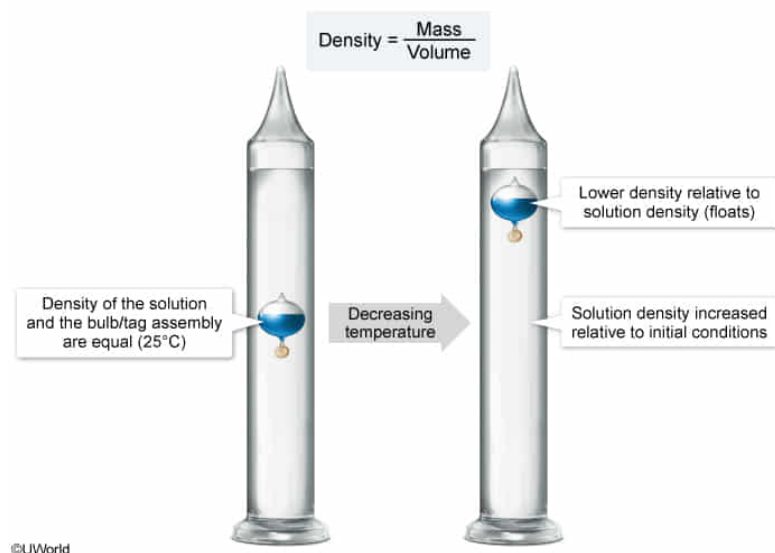


- ☐ A. The cylinder solution density will decrease, and the glass bulb will sink. [19%]
- ☐ B. The cylinder solution density will decrease, and the glass bulb will float. [16%]
- ☐ C. The cylinder solution density will increase, and the glass bulb will sink. [15%]
- ✔ ☒ D. The cylinder solution density will increase, and the glass bulb will float. [48%]

Correct

48% answered correctly

Explanation:



The **density** of a substance is defined as the **mass** it contains per unit of **volume** it occupies. The exact density of a substance, however, can slightly **vary with temperature** due to minute changes in volume because materials tend to **expand when heated** and **contract when cooled**. Materials also physically arrange according to density. For example, when a solid object is placed into a liquid, the object will float if it has a density less than that of the liquid, but it will sink if its density is greater than that of the liquid.

Initially, at 25 °C, the mass-to-volume ratio (ie, density) of the sealed glass bulb exactly matches that of the solution in the cylinder. As the temperature of the solution in the cylinder decreases, the liquid contracts slightly and the density of the cylinder solution increases (ie, same mass occupying a smaller, contracted volume). Because the glass bulb acts like an **isolated system**, its density changes very little compared with that of the cylinder solution. Accordingly, the density of the cooling solution in the cylinder becomes greater than that of the bulb, causing the bulb to begin to float. This is the basis for the function of the **Galileo thermometer**.

(Choices A and B) Most liquids contract and slightly decrease in

volume as the temperature decreases. This causes the mass of the liquid to occupy slightly less space (ie, density increases). Therefore, the density of the solution in the cylinder will *increase* with decreasing temperature.

(Choice C) As the temperature of the cylinder solution decreases, its density increases and becomes greater than the density of the bulb. Objects that are less dense than the cylinder solution will *float* in the solution.

Educational objective:

The density of a substance is defined as the mass it contains per unit of volume that it occupies. The exact density of a substance can vary slightly with temperature due to minute changes in volume caused by the expansion or contraction of materials. Solid objects float in liquids with a higher comparative density and sink in liquids with a lower comparative density.

Time spent:0 sec

Subject:

General Chemistry

QID:401621

System:

Atomic Theory and Chemical Composition